

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q13-0198

Newport Hydro Station 100 kV  
April 01, 2015

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0198. The Interconnection Customer proposes a 142.5 MW solar photovoltaic facility interconnecting at the Newport Hydro Station 100 kV substation in the Ohio area. The request was received on January 21, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 119 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,387,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

The Scott Co to Hooksink 100.0 kV line is overloaded to 125.1% of its 145 MVA emergency rating for the loss of the Scott Co to Burr Oak 100.0 kV. The project contributes 18.7% DFAX (26.6 MW) to this overload.

For the loss of Scott Co to Hooksink 100.0 kV, the Burr Oak to Scott Co 100.0 kV loads to 110.2% of its 144 MVA rating. Project DFAX: 15.0%.

Newport Hydro Station to Winkler 100.0 kV L: 104.7% loading (107 MVA) following the outage of the Newport Hydro Sta to Scott Co 100.0 kV; DFAX = 100.0%.

The Villa Park to Wolfe 230.0 kV line is overloaded to 104.2% of its 768 MVA emergency rating for the loss of the Wolfe to Barrett 500.0 kV. The project contributes 14.9% DFAX (21.3 MW) to this overload.

For the loss of Wolfe to Barrett 500.0 kV Line, the Washington County Cogeneration Facility to Mossyrock 230.0 kV loads to 104.0% of its 618 MVA rating. Project DFAX: 12.8%.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Scott Co to Hooksink 100.0 kV                       | Rebuild line with higher-capacity conductor | \$2,907,000   | 100% (\$2,907,000)    |
| Scott Co Substation to Burr Oak Substation 100.0 kV | Rebuild line with higher-capacity conductor | \$3,052,000   | 100% (\$3,052,000)    |

| Facility                                                      | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------------------------------|----------------------------|---------------|-----------------------|
| Winkler to Newport Hydro Station 100.0 kV L                   | Reconductor line           | \$816,000     | 100% (\$816,000)      |
| Wolfe to Villa Park 230.0 kV                                  | Reconductor line           | \$5,196,000   | 14% (\$726,000)       |
| Washington County Cogeneration Facility to Mossyrock 230.0 kV | Reconductor line           | \$16,703,000  | 17% (\$2,886,000)     |
|                                                               | Total allocated to project |               | \$10,387,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.